Faculty of Engineering and Architecture

Department of Computer Engineering

CEN 302 - Software Engineering

Eren Travel Agency Requirements Specification

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Prepared by:

Ajmi Capo

Anxhela Todaj

Donald Ferati

Ergi Lika

Idon Nurja

Renis Veliu

Uendi Velaj

Received by:

M.Sc Shkumbin Fida

M.Sc Megi Nako

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1. Executive Summary

1.1 Project Overview

The Travel Management System is a web-based platform designed to connect travelers with travel agencies in a unified digital environment. In traditional travel planning, users often rely on multiple websites or direct communication with agencies, which can be time-consuming, inconsistent, and difficult to manage.

This system addresses that problem by providing a centralized platform where travelers can explore, compare, and book travel packages in a single place. At the same time, travel agencies can publish and manage their offers, interact with customers, and monitor bookings more efficiently.

The platform supports key functionalities such as user registration, travel package browsing, reservation management, and feedback, enabling a structured interaction between users and agencies. By consolidating these processes into one system, the platform improves accessibility, reduces complexity in trip planning, and enhances overall communication between stakeholders.

1.2 Purpose and Scope of this Specification

This document defines the functional and non-functional requirements for the Eren Travel Agency Management System. Its purpose is to establish a clear understanding of the system’s features, behavior, and limitations for all stakeholders involved in the project, including developers, designers, and project managers.

In scope:

The scope of this system includes the development of a web-based platform that enables interaction between travelers, travel agencies, and administrators. The system supports the following core functionalities:

- User registration, authentication, and profile management
- Browsing, searching, and filtering travel packages
- Booking and managing travel reservations
- Travel package creation and management by agencies
- Review and rating system for user feedback
- B2B client registration and access to specialized pricing
- Administrative tools for monitoring users, agencies, and platform activity

Out of Scope:

The following features are not included in the current version of the system:

- Direct integration with airline or hotel internal reservation systems (except via third-party APIs)
- Development of native mobile applications (Android/iOS) in this phase
- Implementation of custom payment gateways (external services will be used)
- Advanced AI-based recommendation or personalization systems (planned for future versions)

2. Product/Service Description

In this section, describe the general factors that affect the product and its requirements. This section should contain background information, not state specific requirements (provide the reasons why certain specific requirements are later specified).

2.1 Product Context

The Travel Management System is a web-based application that functions as a centralized platform connecting multiple user roles, including travelers, travel agencies, business clients, and administrators. The system is designed to facilitate interaction between these actors by providing a unified environment for browsing, booking, and managing travel services.

The system is not fully standalone, as it relies on several external services to support its core functionalities. These include:

- **Payment gateways** for processing online transactions
- **Email services** for sending booking confirmations, notifications, and updates
- **Cloud hosting services** for data storage and system availability
- **Authentication services** for secure login and user management

The platform acts as an intermediary between travelers and travel agencies, allowing agencies to publish and manage travel packages while users can search, compare, and book them. By integrating internal system components with external services, the platform ensures efficient communication, reliable transaction processing, and continuous availability.

2.2 User Characteristics

The system is designed to support multiple types of users, each with different roles, levels of experience, and interaction patterns. Understanding these user characteristics helps ensure that the system is designed to meet their needs effectively.

1. Traveler (Client)

- Search for travel packages based on destination, price, or travel dates
- View detailed information about travel offers
- Book travel packages directly through the platform
- Track past and current reservations
- Follow travel agencies to receive updates and promotions
- Leave reviews and ratings for services

2. Travel Agency

- Publish travel packages and promotional offers
- Update or remove travel listings
- Manage bookings made through the platform
- View customer reviews and feedback
- Track the performance of their travel offers

3. Platform Administrator

- Manage users and travel agencies
- Monitor activity across the system
- Review reported issues or inappropriate content
- Generate reports about bookings and platform usage

4. B2B Client (Business User)

- Register using travel agency details and tax ID to access B2B features
- Get verified to use the platform as a business client
- Browse travel packages with special B2B pricing
- Book travel packages at business rate

2.3 Assumptions

User Assumptions

- Users will have access to a stable and reliable internet connection when using the platform.
- Users are expected to have basic digital literacy and familiarity with web applications.
- Travel agencies are capable of managing and updating their own travel packages and content.
- Users and agencies will provide accurate and valid information when interacting with the system.

Technical Assumptions

- The system will be accessed through modern web browsers such as Chrome, Firefox, Edge, or Safari.
- External services (e.g., payment gateways, email systems) will be available and function reliably.
- The platform will operate in a web-based environment without requiring local installation.

Operational Assumptions

- The system will initially support a limited number of languages (e.g., English).
- The number of users and travel agencies will be manageable within the initial system capacity.

2.4 Constraints

1. Technical Constraints:

- The system must be developed using a specific technology stack (e.g., web frameworks and relational databases).
- Hosting infrastructure may limit performance, storage, or scalability in early stages.

2. Security Constraints:

- The system must implement secure authentication and data protection mechanisms.
- Sensitive data must be encrypted and protected from unauthorized access.

3. Operational Constraints:

- The system must maintain logs and audit trails for monitoring and troubleshooting.
- Continuous availability is required, with minimal downtime.

4. Design Constraints:

- The user interface must be responsive and compatible across multiple devices.
- The design should follow standard UI/UX principles for usability.

5. Business Constraints:

- Development must be completed within a limited timeframe and budget.
- Priority is given to core features over advanced functionality.

2.5 Dependencies

External Dependencies

- The system depends on **third-party payment gateways** for processing booking payments and financial transactions.
- It relies on **email services** to send notifications such as booking confirmations, updates, and account-related messages.
- The platform depends on **cloud infrastructure** for hosting, data storage, and system availability.
- Continuous **internet connectivity** is required for both users and system operations.

System Dependencies

- The **authentication module** must be implemented before enabling booking and other user-specific functionalities.
- Reservation and booking features depend on the availability of **travel package data** provided and maintained by travel agencies.

Organizational Dependencies

- The system depends on **travel agencies actively managing and updating their listings**, ensuring that available packages are accurate and up to date.